

GOATUI: Generating Opportunistic Adaptive Tangible User Interfaces through Function Alignment in Augmented Reality

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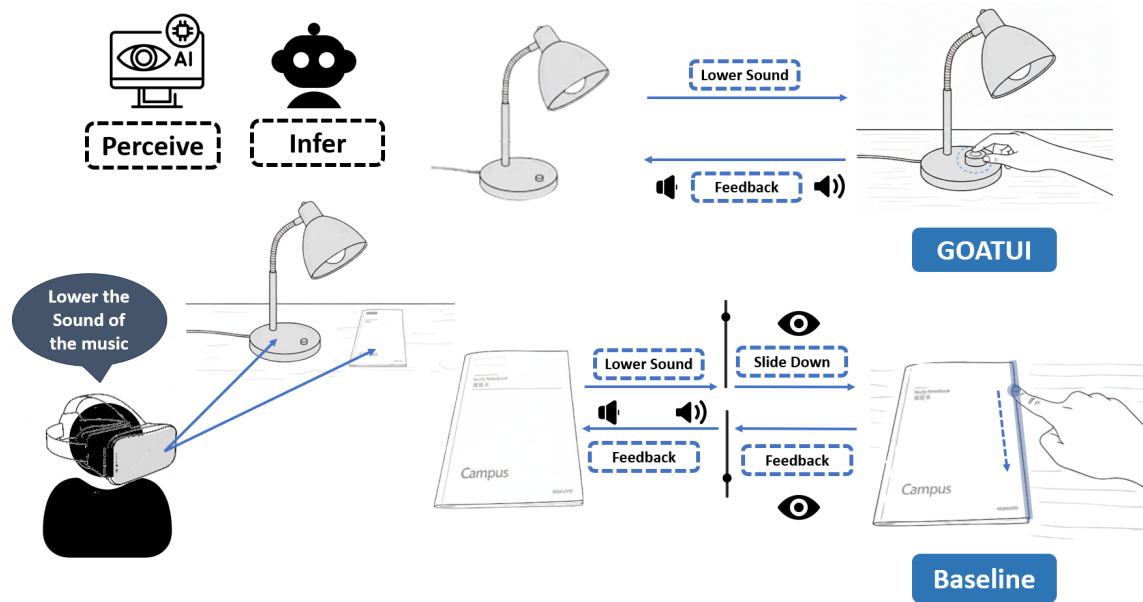


Fig. 1. Our function-aligned approach leverages embodied interactions (Up) vs. Baseline approach utilizes surface-based interactions (Down). Rather than mapping onto the surface of the notebook to imitate the form of the spatial widget, GOATUI directly enables user to rotate the knob of the lamp to adjust the volume providing more intuitive and immersive experience with opportunistic TUI. Enriched by the evolution of AI, our system could understand the environment and automatically infer the appropriate interaction for users.

Opportunistic Tangible User Interfaces leverage everyday objects to provide haptic feedback when users interact with spatial interfaces. Existing approaches of generating Opportunistic TUI rely on manual assignments or user-established adaptation rules and focus mainly on translating surface-based logic to the physical world, overlooking the rich interaction potentials of objects. Building on a formative study, we introduce GOATUI, an AR system that automatically generates TUIs by directly aligning the function of objects' embodied interactions with the goals of digital tasks. The system employs a VLM to perceive objects' interaction potentials within the environment and a vision-based pipeline to define each object's interaction realm. GOATUI leverages LLMs to consider the interactions' functional structure compatibility and functional purpose congruence with tasks to generate suitable TUIs. Our user

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study demonstrates that automatic TUI generation through function alignment reduces mental load and creates more expressive and immersive experiences, increasing users' willingness to engage with TUIs.

CCS Concepts: • **Human-centered computing** → **Interactive systems and tools**; **Mixed / augmented reality**.

Additional Key Words and Phrases: Opportunistic Tangible User Interfaces, Mixed Reality, Everyday Objects

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1 Introduction

Mixed reality (MR) envisions a future where digital and physical worlds merge seamlessly, enabling rich and embodied experiences. Conventional interaction techniques like mid-air hand gestures are often used to interact with virtual content in MR. However, mid-air gestures lack tangibility and tactile feedback, leading to usability challenges [39, 46]. Tangible User Interface [40], especially Opportunistic Tangible User Interface [35], uses physical objects within an existing environment to provide a natural bridge between physical and digital worlds. Prior studies have demonstrated that incorporating physical objects into interaction not only enhances spatial input but also enriches user experience [3, 48].

However, despite years of exploration, TUIs in MR still face from two fundamental limitations. First, they lack flexible automation. Current approaches [15, 21, 33, 34, 36, 37, 42] require users to either manually customize TUIs through authoring tools at real time or establish mapping rules to adapt TUIs pre-defined in a prototype scene to new environments. For example, UbiEdge [21] enables users to directly select the detected edges of a physical object as interactive regions, and manually define both the interaction gestures and their corresponding digital tasks. Such reliance makes the systems difficult to scale to novel environments and entirely new user tasks, often forcing the users back to manual reconfiguration which is very tedious. Second, existing approaches [25, 30, 34, 37, 81, 89] mostly use 2D UI metaphors in which physical affordances are primarily used to replicate surface-based operations such as clicking, dragging, or sliding. For example, WorldKit. [81] supports a very rapid creation of touch-based interfaces on everyday surfaces. Gong et al. [25] conducted an elicitation study demonstrating that surface-based gestures are still primarily used to perform 2D spatial interface tasks. These approaches mainly focus on how to embody these surface-based logics in the physical world, which prevents the users from fully realizing everyday objects' potential as a native interaction paradigm. For instance, when users want to adjust the volume of the music, they will try to find a surface/edge to let them move a slider instead of realizing that a knob could also afford this.

Our work is therefore motivated by the question of **how we could fully utilize daily objects' interaction potentials as tangible UI in AR**. We propose to shift from imitating pre-existing forms to focusing on the user's digital intent or from form alignment to function alignment. By conducting a formative study with 8 participants, we identified the major barriers preventing users from utilizing the full potential of everyday objects as TUIs and formulated three design requirements, including 1) Automated Environment Perception and TUI generation, 2) Function Alignment to User Intent, and 3) Leverage Natural and Embodied Interactions. Guided by these design requirements, we developed GOATUI, a proof-of-concept system that automatically generates and performs as a direct mapping of the deep functional structures between the digital intent and the intuitive interaction with objects in the physical world. To illustrate the applicability of the proposed research prototype, we presented potential use cases in office work,

podcast note-taking, and IoT connectivity. To evaluate the usability and user preferences, we conducted an exploratory within-subject user study with a manual configuration and surface-based gestures as baselines. The result shows that automatic TUI generation through function alignment reduces mental load and creates more expressive and immersive experiences, increasing users' willingness to engage with TUIs.

To conclude, our contributions are as follows:

- We conduct a formative study to identify key barriers preventing user from fully utilizing daily objects' interaction potential.
- We derive design requirements for an automatic AR system that shift the targets of TUI mapping from surface-based gestures to embodied interactions of objects.
- We propose the GOATUI, a proof-of-concept system that auto-generates TUIs by mapping user intents to object interactions through function alignment.
- We provide a comparative evaluation between GOATUI and a TUI generation baseline utilizing surface-based gestures, highlighting GOATUI's enhancement on interaction expressiveness, immersion and willingness to use TUI in the future.

2 Related Work

2.1 Opportunistic Tangible User Interface

Tangible User Interfaces (TUIs), first introduced by Ishii et al. [40], transform digital information into physical forms, enabling users to leverage their natural abilities for manipulating objects when interacting with digital systems. However, current TUI projects [5, 11, 63, 74, 80] are mainly based on customized hardware designs and were developed for specific tasks and contexts, making them difficult to integrate seamlessly into everyday environments. With the advancement in Mixed Reality, the concept has evolved into Opportunistic Tangible User Interface [35], which utilizes geometric shapes of everyday objects as passive haptics and spatial references to enable users to accurately and pervasively control digital functions. Previous works [25, 34, 36, 38, 90] in this area have demonstrated its potential for controlling spatial interfaces. For instance, Gripmarks [36] repurposes hand-held objects into interactive surfaces by rendering corresponding AR contents above the physical objects based on different grasps and object shapes. HoloDesk [38] explored interactions with virtual content co-located on a physical tabletop. UbiDisplay [30] demonstrates how desks, walls, and handheld objects can be repurposed to host virtual UI elements, such as buttons, sliders, and menus.

Despite their contributions, these works primarily focused on identifying suitable geometric features of objects onto which conventional surface-based widgets could be mapped (e.g., any sufficiently flat surface could become a touchscreen, any edge could become a slider). While providing haptic feedback, these works only treat the physical objects as passive surfaces for conventional 2D UI metaphors. As a result, interactions remain largely constrained to surface-based gestures (e.g., tapping, swiping) [25], which only acquire meanings when combined with projected UI widgets. This approach, however, largely overlooks the richer, embodied interaction potentials inherent in the objects themselves. Research [1, 68] in embodied interaction has shown that leveraging the sensorimotor coupling between body, objects, and environment can provide more natural, intuitive, and engaging experiences. For instance, everyday embodied actions such as grasping, rotating, or shaking objects carry intrinsic semantic meanings and could serve as powerful input modalities beyond simple touch [1, 67].

Consequently, a significant gap remains in current research: what happens when we move beyond surface-based gestures and instead harness the full embodied interactions of physical objects to interact with spatial interfaces. This

gap motivates our investigation into how embodied interactions can be systematically mapped onto spatial interfaces to generate appropriate Opportunistic TUIs.

2.2 Adaptive Tangible User Interfaces in Mixed Reality

Traditional TUIs are typically designed with predefined pairings between physical objects and digital functions. While this approach ensures stable interaction, it also constrains the versatility of TUIs, as they are difficult to extend beyond their original tasks and designated objects [23]. To address this challenge, researchers have explored adaptive TUI strategies that emphasize End-User Customization. End-User Customization introduces end-users into the design loop, allowing them to define the mapping between physical interactions and digital functions based on the local contexts.

Early approaches primarily relied on augmenting everyday objects with additional hardware to make them interactive. Marker-based TUIs [12, 16, 20, 21, 28, 45] are a representative example. For instance, ARmarkers [45] enabled users to attach fiducial markers to physical objects and link them to digital assets, effectively turning different objects into AR controllers. By simply swapping markers, users could repurpose objects for different digital functions. Beyond markers, other methods integrated hardware sensors [78, 84] or applied conductive materials [85] to objects to capture user input. While effective, such solutions introduce substantial setup overhead: they require tedious preparation and constrain interactions to the capabilities of the attached hardware.

To reduce setup complexity, subsequent research shifted toward leveraging computer vision techniques. Some of them mainly used RGB-D cameras to track the 6 degrees of freedom (6DoF) manipulation of objects in real time, enabling users to bind TUIs to them and directly manipulate digital functions [15, 23, 69]. For instance, Instant User Interface [15] allowed physical objects with pre-scanned 3D models to be transformed into AR controllers. However, this approach requires that the same physical items must be present across contexts. Otherwise, users have to remap the TUIs. Other systems rely on feature-based tracking of objects rather than tracking the whole object. WorldKit [81] adopts an RGBD projector to detect customized touch interactions on surfaces of everyday objects. UbiEdge [34] developed an authoring tool that enables users to directly select the detected edges of a physical object as interactive regions, and manually define both the interaction gestures and their corresponding digital tasks. These approaches allow users to generalize interactions across objects with similar features. Nevertheless, they still depend on manual authoring and configuration, which becomes cumbersome when users need to interact with multiple TUIs in dynamic contexts.

More recently, researchers have sought to reduce manual effort by introducing rule-based mapping systems that automatically adapt across contexts [13, 14, 33]. For example, AdapTUI [33] employs optimization algorithms to remap TUIs from a prototyped scene to new environments by leveraging geometric features of the objects. Similarly, InteractionAdapt [14] enables user to design a working space and automatically adapt into new environments.

While this reduces repetitive authoring, it still requires initial manual configuration in the prototype scene, thereby increasing cognitive load during setup and limiting scalability to new tasks. As a result, we aim to investigate how automatic TUI generation can reduce user burden, enhance interaction experience, and increase users' willingness to adopt TUIs in AR environments.

2.3 Leveraging Artificial Intelligence in Mixed Reality Interaction

Recent advances in artificial intelligence (AI), particularly in Computer Vision (CV) models and Multimodal Large Language Models (MLLMs) [2, 17, 29, 43, 44, 60, 66, 67, 75–77], have opened up new opportunities for interaction in VR. Modern CV models [55, 59, 62, 70] enable systems to identify and segment objects with increasing precision, from drawing bounding boxes to generating fine-grained masks guided by text prompts. Meanwhile, MLLMs extend beyond

traditional text generation, allowing systems to perceive environments, reason semantically, and generate multimodal content in response to natural language queries.

The reasoning ability of LLMs, grounded in common-sense knowledge and contextual understanding, has already been applied to diverse interactive domains such as code assistance [75], social computing [65, 66], and context-aware recommender systems [8, 47]. By combining CV’s perceptual strength with MLLMs’ grounding and referencing capabilities [19, 82], more intelligent and adaptive MR systems become feasible. For example, Dogan et al. [19] integrated an MLLM into the XR-Objects system, enabling automated recognition of objects and their associated semantic attributes within XR spaces. Similarly, Reality Proxy [82] developed an AI-driven pipeline that extracts structural scene information and generates proxies to facilitate remote object interaction.

Building on these directions, our study extends affordance factors [25] to support more embodied and plausible interactions in MR. We integrate these factors with LLM to infer meaningful user interactions that go beyond surface-level affordance detection.

3 Formative Study

We envision an MR system that generates opportunistic TUIs by leveraging the diverse embodied interaction potentials of everyday objects in the environment. However, a key question remains: **what features do users expect from an AR system to support opportunistic TUI generation that they are willing to use in their daily lives?** To address this question, we conducted a formative study to identify the main barriers that prevent users from utilizing these interaction potentials and concluded with three design goals that align with users’ expectations for TUI generation with the AR system.

3.1 Participants

We recruited 8 participants (6 males, 2 females, age: $M = 21.9$, $SD = 1.73$) between the ages of 20 and 25 from a local university with diverse backgrounds, including computer science, finance, and art. Seven participants had prior VR/AR experience on smartphones or head-mounted devices, while the remaining participant was only familiar with the concepts. Each participant received \$7 for the 60-minute study.

3.2 Study Design and Procedure

To investigate how people naturally use everyday objects as opportunistic TUIs and what support they expect from an opportunistic TUI generation system, we designed an exploratory study across two everyday settings: *working* and *leisure*, which are typical for MR usage [24, 57]. These settings span a range of objects and potential digital tasks, allowing us to observe both conventional and novel opportunistic TUI formats.

The study was conducted in three phases: 1) **Pre-study interview:** Participants first watched an explanatory video introducing the two settings, example digital tasks, and typical opportunistic TUI interactions. 2) **Exploration:** Participants moved through two target settings and designed digital tasks. For each setting, participants were asked to watch three videos showing different scenarios in the setting and design digital tasks they would perform. They were asked to select and use available objects to accomplish the task without the experimenters’ intervention. After each participant finished designing, they would gather to discuss the tasks they designed and how they would like to use TUI to accomplish them. 3) **Think-Aloud and Reflection:** Experimenters walked through each digital task with participants, encouraging them to think aloud about their design choices, discuss possible alternatives, and articulate

their expectations for an intelligent assistant system that supports opportunistic TUI generation. These sessions were audio-recorded with participants' consent.

3.3 Findings

Two authors conducted a thematic analysis [6] on the transcripts through iterative coding and discussion. The analysis revealed three main barriers that hinder users from fully utilizing the diverse interaction potentials in their everyday environment: (1) *Path Dependency on Pre-existing Forms*, (2) *Inattentional Blindness to Native Interaction Potentials*, and (3) *Cognitive Load in Decision-Making*.

3.3.1 Path Dependency on Pre-existing Forms. We found that participants exhibited a strong path dependency when designing tangible interactions for digital tasks. Almost all the participants would subconsciously utilize the surface or edge of an object to replicate the operations they were accustomed to on 2D screens. They articulated that “*When considering a digital task, a concrete 2D UI form (such as a slider or a button) would first come to mind, and thus my physical interactions naturally became an imitation of controlling that virtual UI.*” (P2) Also, the alternatives they proposed were firstly related to different forms of 2D UI they encountered before. This mental model, dominated by pre-existing forms, serves as the primary barrier preventing users from exploring more native interaction possibilities.

3.3.2 Inattentional Blindness to Native Interaction Potentials. Participants often failed to notice some feasible interactions in their environment due to salience bias in object selection. When they were asked to discuss alternative interactions for digital tasks, they would like to select objects that were larger or visually distinct. “*I capture the monitor and the keyboard on the desk at first, so I would like to first select interactions from them.*” (P3) Furthermore, some participants said “*there were plenty of objects in the environment, while each of them contained several interactions. It’s impossible for them to be aware of every possible interaction.*” (P1, P2, P4, P6) Thus, less salient but potentially more functionally appropriate objects were frequently ignored. This suggests that even without path dependency on pre-existing forms, users’ natural attentional mechanisms occluded their perception of potentially superior interaction opportunities.

3.3.3 Cognitive Load in Decision-Making. Even if the user could overcome the previous two barriers and perceive all the feasible interaction possibilities, the step of selecting becomes a burdensome task, especially when they concentrate on their main tasks. P4 noted that “*I want to get the digital task done quickly instead of spending time generating TUIs.*” Similarly, P1 stressed that “*If you generate the TUI for me and tell me how to interact with the TUI, I’d like to use it in my daily life. If not, I would rather perform the interactions I’m familiar with.*” Participants spent more time designing TUIs when asked for alternative interaction and explained why in their interview. They worried that the TUI they generated would not fit in the digital tasks, so they had to make comparisons. Meanwhile, they showed their high expectation for leveraging embodied interactions. P6 noted that “*I was thinking about whether I could directly control the cap of the bottle to adjust the volume of the music during the study. That would be great if the system could help realize it!*”

3.4 Design Requirements

Given the identified three barriers and users’ expectations for the system, we synthesized three design requirements for an intelligent system for opportunistic TUI generation.

- **DR1: Automated Environment Perception and TUI generation.** The system should perceive the physical environment in real-time, identifying objects, their interaction potentials, and contextual properties such as position and interaction realm. Afterward, it should construct an inventory of feasible interactions, applying

common-sense filters to exclude options that are physically impractical or ergonomically infeasible. Finally, the system should automatically generate TUIs from this inventory, reducing the user's mental load in decision-making.

- **DR2: Function Alignment to User Intent.** To reduce path-dependency and decide which interaction to map, we propose to shift from Form Alignment to *Function Alignment*, which directly interprets the user's high-level, original intent and analyzes the functional relevance between intent and available interactions.
- **DR3: Leverage Natural and Embodied Interactions.** Besides conventional interactions, physical objects also possess rich embodied interactions that go beyond surface-based gestures. The TUI generation system should be able to leverage these potentials as the main interaction technique, enabling users to engage with TUIs in more natural ways.

4 Proof-of-Concept System

Based on the design requirements, we implement GOATUI, a proof-of-concept system that could automatically generate opportunistic TUI based on function alignment. The system workflow involves three main steps, which can be categorized as (1) **identify objects and generate interaction realms**, (2) **interpret user intent and generate suitable TUI through function alignment**, (3) **enable users to interact with the TUI**.

4.1 Object Identification and Interaction Realm Generation

To fulfill DR1, we need to identify potential objects and possible interactions. A critical challenge here is that we not only need to identify objects but also localize them and define their corresponding interaction realms. To address this, we developed an integrated vision-based pipeline which was shown in Figure 2

Object Identification. Our system first captures RGB images from the Meta Oculus Quest3 Passthrough Camera API as illustrated in Figure 2 and transmits them to Gemini 2.5 flash¹ which is an advanced vision language model that supports both image understanding and natural language reasoning as the primary perception module. Unlike conventional CNN-based detectors like MediaPipe[62], which are constrained by a fixed set of object categories, the VLM provides open-vocabulary recognition, allowing it to flexibly identify objects beyond pre-defined datasets. Another advantage of using a VLM for object identification is its ability to provide fine-grained descriptions of each object instance. For example, rather than simply labeling an object as a "cup," the VLM can output a more specific annotation such as "a red cup on the left side of the table." These detailed annotations can later be used by a segmentation model to generate separate masks for each object, enabling precise localization and facilitating subsequent interaction design. Furthermore, using VLMs is that they allow users to customize object recognition according to their needs. For example, users can choose to exclude certain objects like fragile items or people in the scene from detection by editing the prompt of the object identification. This flexibility not only enhances user safety and privacy but also reduces computational overhead by limiting the scope of perception to meaningful targets.

Interaction Realm Generation. With the objects identified, our system proceeds to generate interaction realms. These realms are spatially anchored to the objects and designed to capture the area within which interactions can occur. It requires precise 3D position as well as the 3D point clouds of the objects. Instead of obtaining the bounding box and projecting back to 3D using the depth information, and removing the background feature points whose distance with the bounding box is beyond an empirical threshold like UbiEdge, our system decided to directly obtain the mask of

¹<https://gemini.google.com/app>

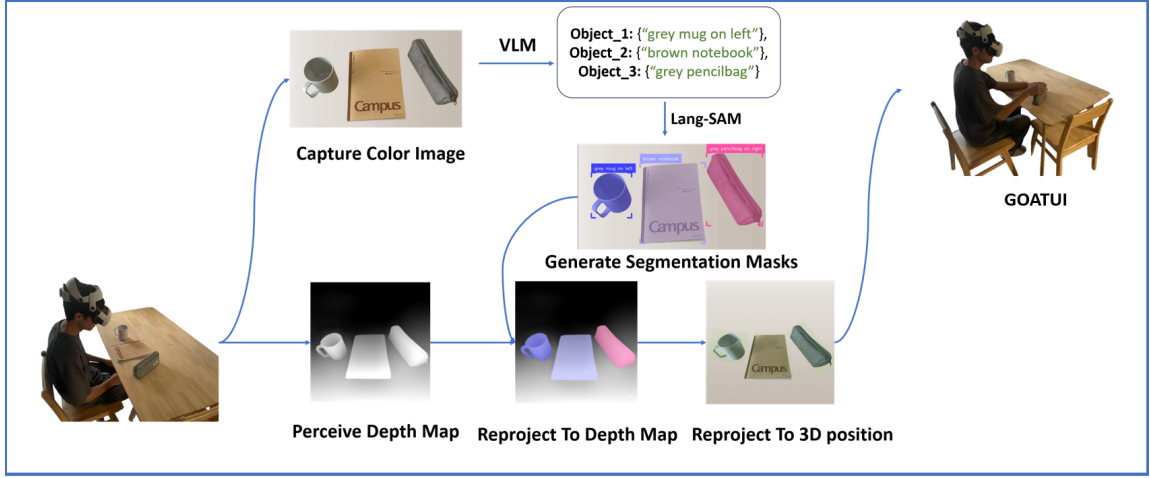


Fig. 2. Pipeline for Object Identification and Interaction Realm Generation

each object. We send the RGB image and the text prompt generated from VLM to Language-Segment Anything², which is an open-source project that was built on the recently released Meta model, Segment Anything Model 2 [70], and the GroundingDINO detection model[59], combining the power of instance segmentation and text prompts to generate masks for specific objects in images. Meanwhile, we access the depth shader at the same frame using the Depth API in Meta Interaction Toolkit, as shown in Figure 2. Due to the spatial offset and differing resolutions between the depth sensor and RGB cameras on the Meta Quest3, the depth and color images have different perspectives. To align them, we first apply a rigid transformation matrix to map depth coordinates into the RGB image frame. Next, we use the object masks to filter and label the corresponding transformed coordinates. Finally, an inverse reprojection matrix is applied to convert these labeled coordinates into 3D world positions, resulting in a spatially aligned point cloud which we utilized to generate the interaction realm for each object.

4.2 TUI Generation

With the objects identified, GOATUI proceeds to generate appropriate TUIs via function alignment. In this subsection, the set of interactions was extended to incorporate more natural and embodied forms, providing a comprehensive reference for grounding interactions with objects. Based on prior works [25], a common functional structure for intents and interactions was established. Subsequently, GOATUI filters out incompatible interactions and examines function congruence, task cohesion, and ergonomics to determine the suitable interaction for a given intent. The pipeline of this step is shown at Figure 4.

Interaction Potentials Inference. While VLMs are capable of directly suggesting object interactions, such outputs often remain open-ended and ambiguous. To address this, GOATUI establishes a more systematic and structured approach for grounding interactions. GOATUI first identifies the affordance factors of physical objects. Prior work has conducted elicitation studies to explore how different affordance factors influence user interactions [25]. However, the interactions they revealed mainly focused on traditional surface-based ones, leaving many natural and embodied interactions unexplored. Building on this foundation, we extended the interaction set to incorporate additional interaction

²<https://github.com/luca-medeiros/lang-segment-anything/tree/main>

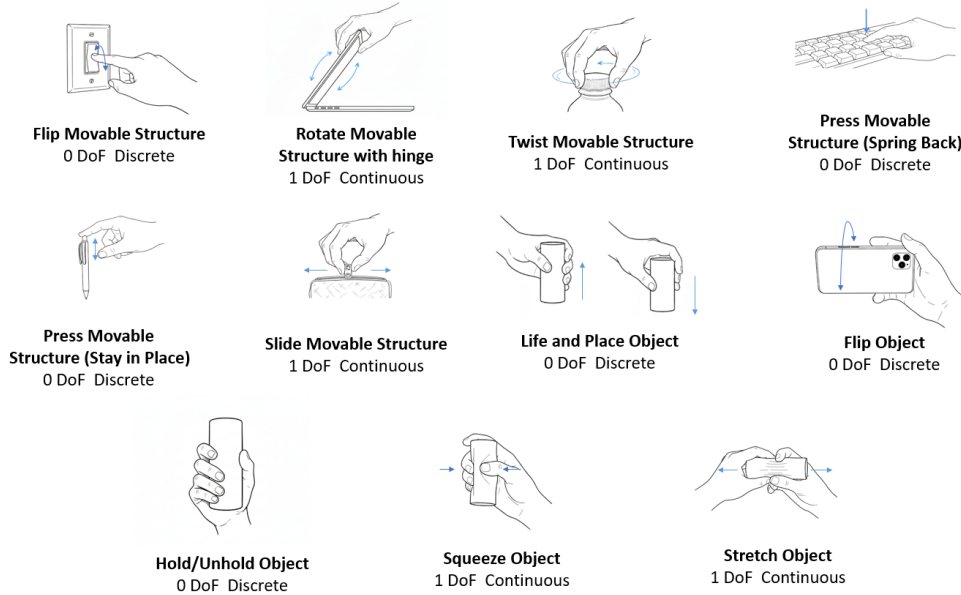


Fig. 3. The Interaction Set for VLM to infer the potential interactions of objects

potentials, particularly those related to affordance factors not extensively explored previously, such as movable structures, grabability, and elasticity, which are critical to many core object functions. To construct the set, we began by examining the 1,202 everyday object categories in the LVIS dataset [27]. After filtering out categories that are not common in daily life, we formed a refined set of 456. For each category, an assessment was made to determine whether it was graspable, elastic, or contained a movable structure. If so, we derived possible interaction potentials with the affordance factor. We then conducted an analysis to synthesize these interactions into an extended interaction set. In Figure 3, we illustrate the set by describing the action and showcasing some examples. Combined with the existing spatial tangible interactions, we formed a more comprehensive interaction set. Consequently, the system can infer corresponding interactions from an object's affordance factors systematically. During the inference, interactions that are impractical, unsafe, or physically infeasible are subsequently filtered out. This ensures that the generated interactions are executable in real-world scenarios.

Function Structure. Previous studies[25] attempted to establish a direct one-to-one mapping between digital tasks and interactions. While straightforward, such mappings are difficult to generalize across the diverse interaction potentials afforded by emerging environments. To move beyond these rigid correspondences, our approach builds on prior works that provide basic characterizations of both interactions[9, 34, 42] and input tasks[9]. From these foundations, we derived a set of functional structures that describe both the requirements of tasks and the capabilities of interactions. This enables the system to systematically filter out unapplicable interactions based on its structural compatibility. Specifically, our functional structure is defined along two key dimensions: (1) Input Degree of Freedom

(DoF). The first structure our system considered was the Input Degree of Freedom derived from Buxton’s input taxonomy, which categorized tasks based on the number of continuous dimensions to be controlled [9]. For physical interactions, it captures the number of independent spatial dimensions that an action can express [25]. (2) Flow: Inspired by another dimension in Buxton’s taxonomy[9], this categorization distinguishes tasks and interactions as either discrete or continuous[9]. We further interpreted each interaction in terms of these functional structures extended in the last step and illustrated them in Figure 3.

Function Alignment Our system aims to directly interpret users’ high-level intents, bypassing pre-existing forms. These intents are input in text format and can range from explicit commands, such as “I want to adjust the volume,” to more implicit statements like “the sound is too loud.” We use an LLM (Gemini 2.5 flash) in a few-shot prompting setup to parse intents into structured functional descriptions which is . Specifically, the LLM outputs both a canonical task label (e.g., adjust volume) and functional structure (e.g., DoF: 1D, Flow: continuous). While this provides a clear requirement, a significant challenge remains: selecting the suitable interactions from the multiple possibilities within the environment. To address this, our system employs a multi-stage filtering and ranking process to identify suitable candidates.

GOATUI first applies a computational filter based on the task’s functional structure. Norman’s theory about the gulf of execution [64] describes the gap between user intention and the actions the system allows them to do, where a wide gulf means the user struggles to figure out how to use the interface to achieve their goal. Therefore, to narrow this gulf, we systematically evaluate and filter out incompatible interactions. Specifically: (1) Any physical interaction with a lower Degree of Freedom (DoF) than required by the task is removed since adequate DoF is essential for tangible interaction [71, 83]. (2) Discrete inputs can only be compatible with discrete outputs since most of them cannot afford continuous output. This initial filtering ensures that all remaining candidates are structurally applicable. After the initial structural compatibility check, the system performs a deeper analysis to evaluate the **Function Purpose Congruence** between the task and the remaining candidate interactions. This criterion is inspired by Norman’s notion about Natural Mapping[64], which articulates that the relationship between the controller and its effect should be natural and intuitive, allowing users to easily understand its operation without proactive learning. Our system operationalizes this principle using a three-tier search: (1) *Direct Congruence*: Direct Congruence means that when a physical interaction’s inherent real-world purpose is directly analogous to the digital task, users could leverage their prior knowledge of interacting with the physical world to effortlessly understand the usage of the TUI, resulting in a seamless interaction experience. Our system will first compare the similarity of the required function and the inherent function inferred during the perception step. If there exists the same, corresponding interactions will be selected as alternatives. (2) *Metaphorical Congruence*: If no directly congruent interactions are found, the system broadens its search to identify candidates with Metaphorical Congruence. Adopting Fishkin’s definition, this refers to where the association of a system effect of a user action is analogous to the real-world effect of similar actions[22]. Even though the interaction may not afford a core purpose directly analogous to the digital task, it is a common, sensible interaction with the object that users could intuitively perform according to the embodied knowledge they have. (3) *No Congruence*: In the event that no congruent interactions are identified, the system establishes a final candidate pool. This pool includes any remaining structurally valid but non-congruent tangible interactions, alongside a set of well-established, traditional surface-based gestures[25]. Afterwards, the candidate interactions identified through the congruence analysis are ranked by contextual appropriateness. In our prototype, we apply two heuristics: (1) *Task cohesion*: The system prioritizes interactions that can be harmoniously integrated into the user’s primary workflow with minimal disruption. (2) *Ergonomics*: Interactions that are physically closer or require less effort are preferred. This comprehensive, multi-stage process allows GOATUI to

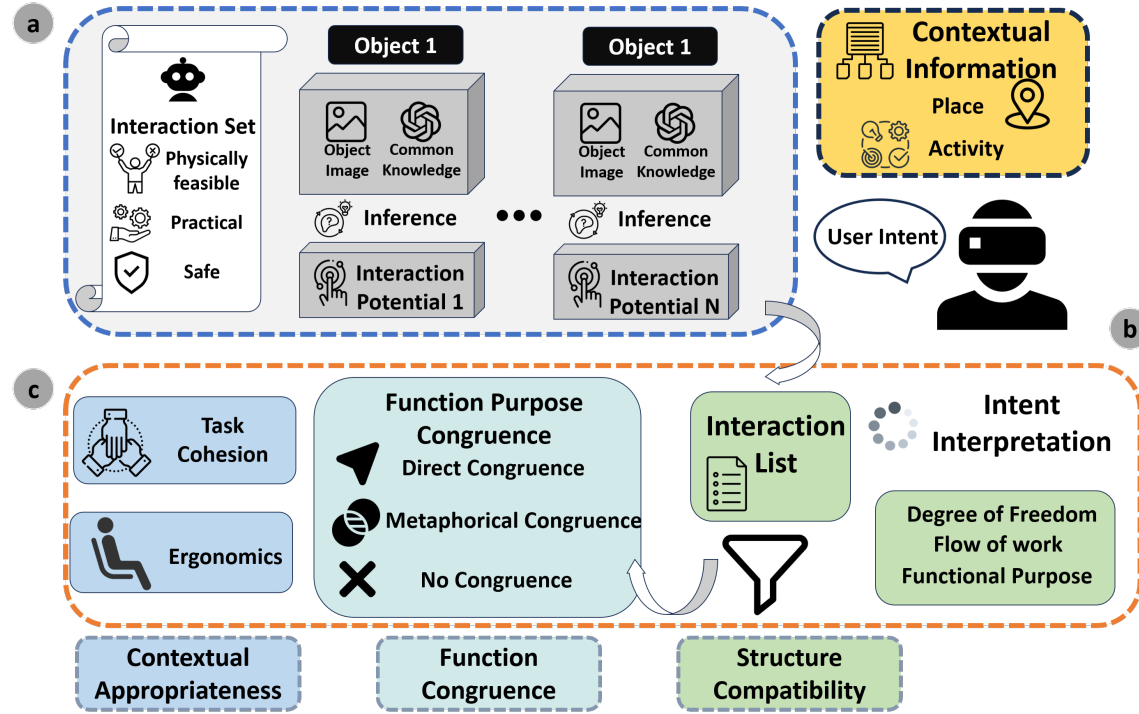


Fig. 4. Pipeline for Opportunistic TUI Generation. (a) GOATUI leverages VLM to infer feasible interactions of each object. (b) GOATUI leverages a LLM to interpret user intent into Functional Structure and Functional Purpose. (c) Function Alignment: Interaction list goes through three steps: (1) Structure Compatibility, (2) Function Congruence, (3) Contextual Appropriateness

systematically evaluate all potential interactions, ultimately recommending the one that is not only usable and intuitive but also appropriate for the user's context.

4.3 Interaction with TUI

To enable users to interact with the TUIs generated in the previous stage, we implemented a gesture-based execution layer using the Meta Interaction SDK³. In our prototype, gestures were defined through the SDK's customization interface rather than trained end-to-end, allowing us to rapidly prototype reliable interactions. To minimize misinterpretation, we created an *interaction realm* for each recognized object. Combined with the Quest 3's built-in hand tracking, the system activates the recognizer only when the user directly engages with the corresponding physical object. Furthermore, the recognizer is constrained to detect only the gestures associated with that specific TUI, which prevents accidental activations of other mappings. This design decision prioritizes robustness and clarity in our proof-of-concept prototype. While our current implementation relies on predefined gestures, future work could extend this layer with adaptive or multimodal recognition to support more generalizable interactions.

³<https://developers.meta.com/horizon/documentation/unity/unity-isdk-building-hand-pose-recognizer>

4.4 Implementation

We use Unity and Meta Interaction Toolkit to bring the necessary components for these steps to build our system. We use the Meta Oculus Quest 3. The headset is primarily responsible for immersive visualization of digital content, as well as tracking of the user’s head and hands, and collecting RGB data and depth data from its two built-in Color cameras and a depth sensor, respectively. Although Meta and Unity provide toolkits such as Senti to run AI models directly on Quest 3, the device’s hardware limitations make it infeasible to deploy large and high-precision models on the headset itself. To overcome the hardware constraints, we establish a WebSocket connection to an external laptop equipped with a high-performance GPU. The Quest 3 then streams real-time RGB and depth data to the laptop to enable object segmentation. For higher-level semantic reasoning tasks, the Quest 3 also communicates with a cloud-based VLM. Requests are sent from the headset and the results are integrated with the perception pipeline to support embodied interaction design. This division enables our system to leverage Quest 3’s mobility and tracking capabilities, external GPU computation for precise perception, and cloud-based VLMs for semantic reasoning, together ensuring reliable and adaptive opportunistic TUI generation.

5 Applications

Through Function Alignment, we envision GOATUI to enrich the choices of TUI across a variety of real-world applications. To demonstrate, we developed three proof-of-concept applications: Office Working, Podcast Note-taking and IoT Connectivity to highlight the value of GOATUI and its potential use cases.

5.1 Office Working

Figure 6 illustrates an example of the workday of Nick, a software engineer, which begins with a cup of coffee. When he sits down at his desk, GOATUI activates and quickly perceives his surrounding environment: a coffee cup, a closed laptop, and a smartphone, generating a set of interactions with these objects. The system first proactively reminds Nick to check unread emails related to his schedule for the day (Figure 5a). In response to this task, GOATUI maps the interaction of "opening the laptop" to the command "display email list." When Nick opens his laptop, several unread emails appear as virtual elements above the screen. Then he notices an urgent email from his boss, GOATUI leverages the coffee cup he is already holding to enable him to rotate the cup to scroll through the emails (Figure 5b). Once he has selected the target email, the action of placing the cup back on the desk is interpreted by the system as the "confirm and open" command (Figure 5c). Upon finding that they will have a meeting in 15minutes, the system suggests he update his online status. For this, GOATUI assigns the action of "flipping the phone" as the TUI for "changing work status" (Figure 5d). After 15 minutes, Nick receives the meeting invitation from his boss, and the system tells Nick he can press the button on one side of the phone to accept it (Figure 5e). This scenario fully illustrates how GOATUI integrates into a user’s workflow as an intelligent assistant. This application shows the potential of the system to understand the user’s potential intent at different stages.

5.2 Podcast Note-taking

GOATUI enables user to proactively input their intents to generate TUIs for them. To demonstrate that, we developed another example of podcast note-taking (Figure 6a). A user, situated at a desk with a notebook, pen, lamp, and pencil bag, expresses the intent to "listen to an HCI podcast and take notes." The system intelligently designates that the physical interaction of opening the notebook serves as a natural trigger to begin the podcast (Figure 6b). In case the

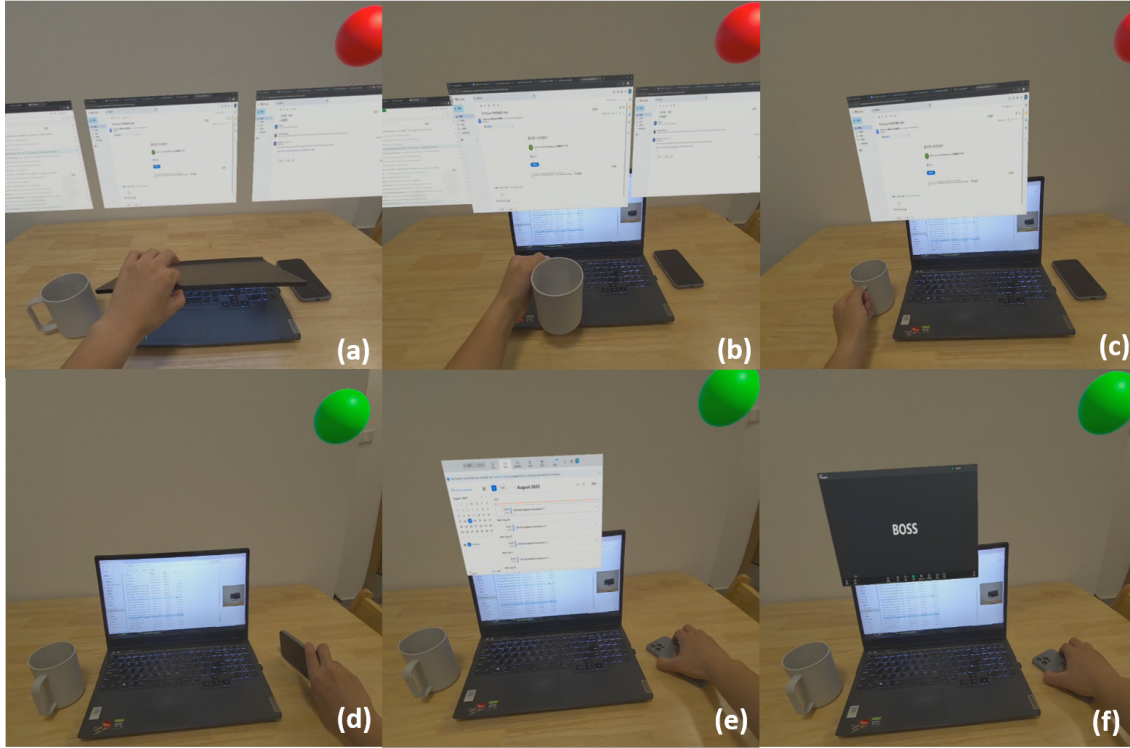


Fig. 5. Office Working scenario: GOATUI predicts the user's potential intent and leverages nearby objects to support email (a) checking, (b) navigating, (c) confirmation, (d) status update, and (e) meeting acceptance.

user wishes to change episodes, flipping the pages of the notebook allows them to browse (Figure 6c). A command like "the sound is too loud" maps volume control to the lamp's rotary knob (Figure 6d). Similarly, a request to "skip the introduction" maps the seeking function to sliding the pencil bag's zipper (Figure 6e). Every time a user wants to make some notes down on the notebook, he picks up the pen and the podcast will consequently be stopped (Figure 6f). This scenario illustrates how GOATUI would intelligently interpret the proactive user's intent to generate appropriate TUIs.

5.3 IoT Connectivity

We also envision potential applications that extend to the control of real-world connected devices (e.g., IoT) so that everyday objects could serve as tangible proxies for remote devices. For example, rotating a bottle cap could adjust the temperature of an air-conditioner or scrolling the wheel of a mouse could modulate the brightness of the light. Such connectivity scenarios illustrate the broader design space that function alignment opens up, particularly in bridging opportunistic TUIs with ubiquitous computing environments.

6 User Study

We develop GOATUI as a technology probe to explore how natural and embodied interactions with everyday objects might influence user experience with TUI. We conducted a user study (N=12) to experience GOATUI as well as manual

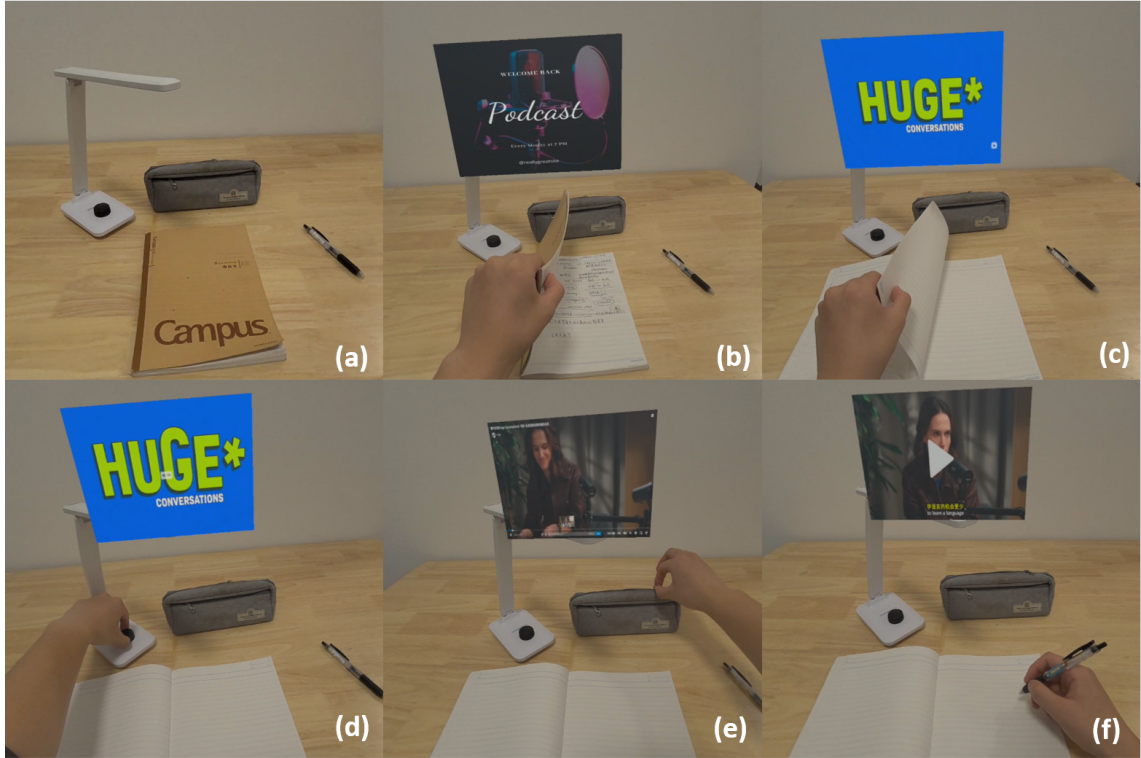


Fig. 6. Podcast Note-taking Scenario: GOATUI receives proactive user intent and leverage (a) nearby objects to support (b) podcasting activating, (c) podcast switching, (d) volume adjusting, (e) progress adjusting, (f) podcast pausing.

configuration and a baseline utilizing surface-based gestures, which was based on the affordance-based object selector scheme from previous work [25]. This table is shown in This baseline is built upon the same pipeline as GOATUI except for the mapping step. The baseline can also automatically generate opportunistic TUI for user intent to ensure that we could evaluate the influence of embodied interactions by comparing them. We chose this baseline, hoping to give participants a reference for current TUI types to articulate richer and more insightful feedback. In the study, participants were asked to perform a number of tasks with TUI in simulated working and at-home scenarios, and provide feedback on their experience with each technique through interviews.

6.1 Participants and Apparatus

We recruited 12 participants (5 female, 7 male) between the ages of 20-28 from several universities. 10 out of 12 had used AR or VR applications on smartphones, tablets, or head-mounted devices, while the remaining two had heard about AR and VR. Each participant was rewarded \$10 for a 60-minute session. The study was conducted in a controlled indoor environment using a Meta Quest 3.

6.2 Task & Procedure

6.2.1 Introduction and Training. Participants were first given a brief introduction of the motivation of our study as well as the concepts of Tangible UI. They then signed the consent forms, filled out a demographic survey and received a tutorial on the Meta Quest3's basic interaction. The experimenter then walked through the interactions potentials of objects provided in experiments and basic surface-based gestures with the participants. They were asked to perform the gestures in a debugging scenario provided by Meta to ensure their gestures could be recognized by the application. Participants were encouraged to ask questions at any time, and training phase continued until they felt confident performing the gestures.

6.2.2 Tasks. We designed two scenarios consisting of an simulated office working scenario (Figure 5) and an at-home scenario (Figure 6) which adapted from section5 to help participants experience both tangibles UIs utilizing embodied interactions with everyday objects and barely surface-based gestures. (1) **Task-1:** Participants were asked to open the mailbox when they arrive at the desk, select the email sent by his boss and open it. After reading the content of the email, they were asked to change their working status and accept the meeting when received the invitation. (2) **Task-2:** Participants were tasked with listening to the podcast and make notes of the key points. They were first asked to start the podcast and switch to the specific one. Then participants would adjust the volume until they were satisfied and jump over the music part of the podcast. When they heard some valuable information, they would stop the podcast to make some notes. For each task, participants first conducted a manual configuration utilizing the interactions introduced in the training phase to generate TUIs. Then they will experience both conditions of GOATUI and the baseline. The ordering of both conditions and scenarios (A, B) were counterbalanced between participants to prevent ordering effects.

6.2.3 Semi-Structured Interview and Questionnaire. At the end of the session, each participant completed two 5-scale Likert-type questionnaires, NASA-TLX [31], SUS [7], and UEQ-S [50]. We then asked the participants to take a rest for five minutes and recall the experiments. After the break, the experimenter conducted a semi-structured interview to further gather qualitative insights regarding the benefits, limitations for improving, and potential application scenarios. All interviews were audio-recorded for post-study analysis.

6.3 Results

All 12 participants successfully completed both tasks using our system and the baseline. The qualitative results from the session are shown in Figure 7 and Figure 8. We conducted a series of Wilcoxon signed-rank tests to analyze the scores. To analyze the qualitative data, three authors transcribed all semi-structured interviews and independently conducted descriptive coding. The researchers then met to compare codes and resolve disagreements through discussion.

6.3.1 Overall System Usability. We did a one-tailed single sample Wilcoxon signed-rank tests against the neutral midpoint of 3 for all items in the questionnaire assessing the automatic system. Results showed that all features were rated significantly higher than the neutral midpoint ($p < .05$). Overall, the system achieved an average usability score of 74.38 out of 100 ($SD = 8.10$), indicating that the system is both promising and highly usable. Participants reported that the system was enjoyable to use (Q3: $AVG = 4.00$, $SD = 0.60$). "I have used a VR headset before but the system offer a completely different experience which I enjoyed a lot. I think it could be the future of Mixed Reality, especially with the evolution of AI." (P3) Besides, participants (e.g. P2, P4, P12) consistently emphasized that an automatic system was very helpful (Q4: $AVG = 3.83$, $SD = 0.72$) for leveraging the interaction potentials of everyday objects as tangible inputs. "I like the way everything is arranged well for me automatically. It feels like having a full-time assistant always

ready for me." (P4) They praised the system's convenience compared to manually assigning TUIs, which they described as time-consuming and mentally demanding (Q6: AVG = 4.08, SD = 0.52). *"The procedure of manual configuration is both tedious and cognitively demanding. Having a system to help me understand my environment and plan is very convenient, especially when dealing with multiple intents during my activity."* (P2). Moreover, participants agreed that the system was able to understand the environment efficiently (Q1: AVG = 4.17, SD = 0.72). *"Without the system I need to anticipate all possible interactions in advance, I have no worry about this with the help of the system."* (P4) Participants also acknowledged that directly expressing user intent provided more freedom and flexibility for both the user and the system (Q2: AVG = 4.08, SD = 0.79). *"I no longer need to think about the exact UI forms for different applications, but only about what I want to with the app. I think it could become the next version of Voice Control."* (P7) Although participants generally indicate that they could quickly learn and understand the TUI generated by the system (Q5: AVG = 3.83, SD = 0.94), some (P5, P9) pointed out the need for additional guidance. *"It will also raise some learning demand and I would like an explicit tutorial or indicator to avoid the legacy bias caused by previous experience with TUIs."* (P5) We will discuss this concern further in Section 7.

6.4 Comparison between GOATUI with baseline

In general, the overall score of UEQ-S shows that users had a better experience using GOATUI (AVG = 5.86, SD = 0.67) compared to the baseline (AVG = 3.96, SD = 0.98) $p < 0.05$. Specifically, the hedonic quality of GOATUI (AVG = 6.23, SD = 0.68) is significantly higher than the baseline (AVG = 3.31, SD = 1.58) $p < 0.05$. While pragmatic quality has no significant difference between GOATUI (AVG = 5.5, SD = 0.73) and the baseline (AVG = 4.60, SD = 1.28) $p > 0.05$. *"I'm very familiar with surface-based gestures. Although the embodied interactions are also intuitive, but after all this kind of interaction is too novel to be widely used for now."* (P5) However, most of the participants are willing to use GOATUI (AVG = 4.38, SD = 0.74) than the baseline (AVG = 3.25, SD = 1.04) $p < 0.05$. P5 further noted that *"I would imagine that the system will be much more pragmatic when they get familiar to this technique."* GOATUI (AVG = 3.88, SD = 0.83) is significantly more intuitive than baseline (AVG = 3.125, SD = 0.83) $p < 0.05$ is one of the reasons. Some participants mentioned that *"Since we don't have specific UI widgets in the experiments, it will take some time for me to figure out which part of the surface or which edge to perform the surface-based gestures. However, embodied interaction with objects don't require this for the interaction with object is very clear and intuitive."* (P1) They also thought it was the reason why they need more effort with the baseline (AVG = 4.13, SD = 1.64) than with GOATUI (AVG = 2.88, SD = 1.36) $p < 0.05$. As a design foundation of our system, the embodied interactions received a highly agreement of enhancing the sense of control and immersion (AVG = 4.00, SD = 0.053) than the baseline (AVG = 2.13, SD = 0.099) $p < 0.05$. Participants contrasted this directly with the feeling of interacting with surface-based interfaces in the baseline. For instance, P2, who has high standards for interaction quality, articulated that with our system, he preferred the sense of *"directly interacting with the physical object"* rather than merely *"controlling a digital proxy on the screen"*. Similarly, P6 described the experience as akin to *"using a joystick or mouse to play a game"* from which she derived a strong sense of embodiment that she emphasized was absent when *"simply moving fingers on a surface"*. They also stressed that the immersion of embodied interaction enabled them to perform (AVG = 4.38, SD = 1.51) better than the baseline (AVG = 3.25, SD = 1.58) $p < 0.05$. Moreover, participants emphasized that the system enabled more interesting and expressive forms of interaction (AVG = 4.125, SD = 0.35) than baseline (AVG = 2.5, SD = 1.07) $p < 0.05$. *"When I do the experiments under the baseline condition, I was repetitively performing gestures like tapping, dragging and swiping. However, interacting with the real-objects offer diverse gestures which is much more expressive."* (P7) Also, participants agreed that the TUIs generated by the system were closely related to their intents (AVG = 4.38, SD = 0.74) than baseline (AVG = 3.13, SD = 1.36) $p < 0.05$. *"Rotating the knob*

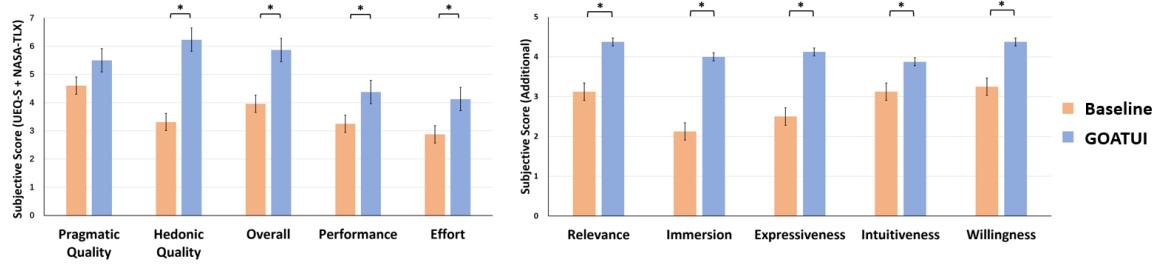


Fig. 7. Summary of evaluation results. (a) Subjective scores for statements from UEQ-S questionnaire and NASA-TLX questionnaire. (b) Subjective scores for additional Likert-type questionnaire of both GOATUI and Baseline

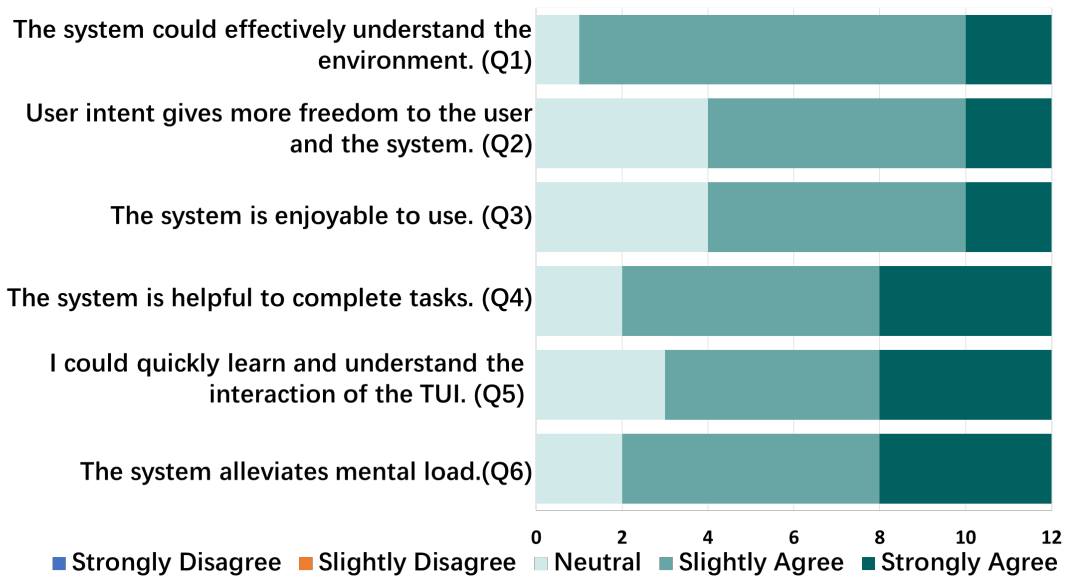


Fig. 8. Summary of evaluation results. Subjective scores for additional Likert-type questionnaire of GOATUI

of the lamp impress me the most which makes me feel like I was directly controlling the volume of a speaker." (P2) While some participants (P6, P11) who gave higher score on baseline mentioned that "Prior experiences on 2D screen still have strong influence on my interaction with TUI."

7 Discussion

Context-aware Augmented Reality aims to seamlessly blend virtual elements by automatically adapting them based on users' needs, context, and environment [26, 56, 58]. Recent research on Opportunistic TUIs leverages everyday objects to provide affordances for virtual interactions. Although they offer a way to embed virtual elements, they mainly focus on matching the form of the spatial interface to move 2D widgets to surfaces or edges [34] which fails to utilize diverse interaction potentials conveyed by everyday objects themselves which enable a more seamless and adaptive

experience [73]. We contribute to GOATUI, a novel assistant system that perceives daily objects' interaction potentials and aligns their function with user intents to generate appropriate TUI to facilitate the user experience. By proactively analyzing the environment and user intents, GOATUI perceives interactions may overlooked, bridging the gap from "what object can I use as TUI" to "which object I want to use as TUI". This approach minimizes the mental load and offer diverse and intuitive interaction possibilities, enhancing user engagement with MR and fostering their willingness to utilize TUIs.

Willingness to Leverage TUI A critical insight from our study concerns how users' willingness to adopt TUI arises. Some participants, despite their XR experience, initially questioned the necessity of TUIs since currently mid-air gestures and eye-hand coordination are the mainstream of interacting with virtual contents. As P1 noted, *"I feel no difference between eye-hand interaction and interacting with UIs on the surface of my desk."* (P1) Another barrier concerned the effort required to generate TUIs. Current approaches often rely on authoring tools that allow users to manually bind physical objects to digital functions [34, 51, 88, 89]. While such tools support user-defined mappings, they are cumbersome in daily life, particularly when users change environments or wish to leverage multiple TUIs. The additional steps of authoring discourage users to proactively use it. Prior research also suggests that the future of MR should expand to understand the contextual environment rather than remain limited to conventional input channels [73]. Our study therefore reveals that the motivation to use TUIs stems from fundamentally new forms of interaction experience. Participants valued TUIs not as replicas of digital screens, but as an opportunity for the development of MR. Therefore, the system's ability to reveal unexpected interaction potentials proved to be a powerful motivator. Furthermore, participants (P1, P10) also noted that *"Embodied interactions with daily objects lower the cost from transferring from physical world to digital world."* Traditional form alignment maps 2D UI widgets to surfaces or edges, users often require explicit visual cues to ensure correct interaction while embodied interactions alleviate this problem. We therefore envision future research to explore how such embodied interactions might lower the cost transferring between digital and physical world.

Integrating GOATUI with Multimodal Input In our study, user intents were manually input by the experimentors. One question emerged during post-study interviews was how to input user intent into the system in future life. Prior works leveraged multimodal input like eye gaze [10, 79], voice input [41, 49, 52, 86] as well as contextual information [54, 72] like location and user profile. Therefore, some participants (P5, P9) expressed expectations for voice interaction as a proactive input method. They acknowledged that relying exclusively on voice control can be tedious, especially for frequently used functions that require repeatedly articulating the same commands. However, they saw value in combining voice with tangible interaction: using voice to quickly generate a specific TUI, and then manipulating the physical object to refine the action. This hybrid process allows users to begin with high-level intent expression and then engage in a more gradual, exploratory refinement through embodied interaction. Furthermore, P4 proposed that in some areas where voice input is inapplicable, eye gaze could be used to trigger the system. Previous work has proved the usability of eye gaze as a trigger [87]. P4 suggested that user could directly perform a glance at the widget he wants to interact [61] and the system generate TUI within the environment. Meanwhile, P7 highlighted the value of predictive assistance, suggesting that the system should infer the user's likely intent without requiring explicit requests. For example, when cooking, after the user places all ingredients into the pot, the system could automatically predict that the next step is setting a timer. In this case, the pot itself could serve as the tangible control to adjust the cooking time. He emphasized that such predictive behavior would be particularly useful for one-time or time-critical intentions. We envision that future research to discover how to enable GOATUI to capture user intent as a predictive assistant. We hope through these studies to develop how to fully integrate not only GOATUI but also MR into daily life.

Learning Curve For Interactions While all the participants praised that the automation system successfully removed the mental load of the *“boring and repetitive pre-definition process”*, they also identified a new learning curve associated with adapting to new user intents and contexts. Due to the variation of the surroundings, GOATUI may generate different TUIs for the same intent and users have to overcome the influence of prior experience to learn how to use the new TUI. P1 described this learning process as *“learning to play a new game using a controller. Though the buttons remain the same, the function of them are different.”* (P3). Our study found that users could quickly learn how to use the generated TUI but they have different opinions towards this learning curve. With 4 out of 12 participants propose their worries about the learning curve. *“For some urgent tasks, I am not sure whether I could quickly understand the generated TUI.”* (P9) Prior research [64] investigated that in task-oriented contexts which need high efficiency, this learning process would increase frustration rather than enable productivity without explicit tutorials or indicators. Conversely, in exploratory contexts, the learning process was experienced as enjoyable and motivating [18, 32]. P4 and P6 noted that *“Learning how to interact with new TUIs offered me a novel, interesting and exploratory experience.”* These insights suggest future research could adaptively balance guidance and freedom, minimizing unnecessary burden in efficiency-driven scenarios while preserving opportunities for curiosity-driven exploration.

Sustaining human agency A recurring theme in participant feedback was the importance of sustaining human agency. While our current implementation relies on VLM and CV models with Quest 3 sensors to perceive the environment, participants noted that system-driven perception inevitably introduces errors like overlooking objects due to sensor limitations. Although improving sensing hardware could mitigate such issues, participants emphasized that a more robust solution is to actively involve users in refining the system’s perception. For example, if an object is missed, users should be able to manually indicate its presence through raycasting selection enabled by MRUK. Conversely, when users do not wish to include certain objects or interactions, the system should provide intuitive mechanisms to exclude them and regenerate TUIs accordingly. Beyond these immediate corrections, participants also expressed a desire for the system to learn and remember their personal preferences over time, enabling smoother adaptation across contexts. We anticipated that GOATUI could balance automation with opportunities for user intervention and preference learning.

Limitation and Future Work Our study has several limitations that open avenues for future work. First, In our current implementation, object identification and interaction realm generation are triggered by a button press on the controller rather than performed continuously, due to the latency introduced by the VLM and Lang-SAM models. At present, the interaction realm of each object is represented by a bounding box that encloses the entire object. Since the depth sensor cannot capture occluded regions, the resulting point clouds are incomplete, making it infeasible to construct a precise convex hull that tightly conforms to the object’s geometry. Future research should explore how to sense and generate interaction realm in real time using light-weight models as well as use completion models to complement the convex hull to lower the possibility of accident interaction. a fundamental challenge lies in the recognition of user interactions with objects. Although we attempted to define interaction gestures using the Meta Interaction Toolkit, the diversity of user gestures rendered this approach not enough. We also experimented with uploading video streams into a VLM for recognition, similar to Satori [53], but the resulting latency was too high to support the instant responsiveness expected by users. We envision that future work could leverage the interaction realm introduced by our system to filter potential object affordances, thereby enabling faster and more practical interaction recognition on devices such as Meta Quest 3 or AVP. Third, concerns about the physical impact on objects during interaction remain a longstanding issue for TUIs. Participants worried that their physical interaction would influence the physical world which may cause negative effect. Future research should carefully consider take this into consideration. Additionally, we use LLM to accomplish

the inference of the interaction set and TUI generation. Even though we've provided reference to enhance LLM's ability to reduce hallucination, inaccuracies of LLM generated-contents still occur [4]. Finally, our evaluation was conducted in a controlled environment with a limited set of tasks and objects. This raises questions about the ecological validity and generalizability of our findings. Future studies should examine how the system performs in more complex and diverse real-world environments such as homes, offices, and public spaces.

8 Conclusion

In this paper, we introduced GOATUI, a novel proof-of-concept AR system to automatically generate opportunistic TUI for user intents. Its key design principle is function alignment which is to shift the targets of TUI mapping from surface-based gestures to embodied interactions of objects to match the function of user intent. By decoupling user intent from spatial UI widget, GOATUI enables users to interact more immersively and intuitively with object through embodied interactions. With the evolution of MLLM and Computer Vision, GOATUI further support proactively understanding the environment and generating appropriate opportunistic TUI for users. Through a user study comparing GOATUI with manual configuration and a baseline built upon surface-based gestures, we demonstrate that GOATUI not only alleviates the need for manual configuration, but also generates expressive and intuitive opportunistic TUIs enabling the interaction more immersive. Our post-study interview highlights GOATUI's potential to enhance users' willingness to use opportunistic TUI and to foster the development of MR.

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```
{
  You are an expert in Human-Computer Interaction, specializing in parsing user intents for tangible interfaces. Your task is to analyze a user's natural language command and translate it into a structured task definition based on its functional properties.
  The functional properties are:
  DoF (Degree of Freedom): The number of continuous dimensions to be controlled (R0, ID, 2D, 3D, etc.).
  Flow: Whether the interaction is "Discrete" (a single action, like a click or toggle) or "Continuous" (an ongoing adjustment, like dragging a slider).
  Provide the output as a single JSON object with the keys "task_name", "DoF", and "Flow". If the input contain several tasks, separate each task and give output respectively.

  Example
  User Input: "the sound is too loud"
  Output: {"task_name": "adjust_volume", "DoF": "ID", "Flow": "Continuous"}
  User Input: "turn on the light"
  Output: {"task_name": "toggle_light", "DoF": "R0", "Flow": "Discrete"}
  User Input: "the sound is too loud, I also want to turn on the light"
  Output: {"task_name": "adjust_volume", "DoF": "ID", "Flow": "Continuous"}, {"task_name": "toggle_light", "DoF": "R0", "Flow": "Discrete"}
  User Input: [Input your intent here
}
```

Fig. 9. Prompt For Understanding User Intent

```
You are an expert system for Human-Computer Interaction, specializing in Tangible User Interfaces (TUIs). Your goal is to analyze a LIST of candidate interactions for a digital task, classify each one, and then select the SINGLE BEST interaction based on the "Purpose-Action Congruence" framework and a prioritized decision logic.
--
You must strictly follow this custom framework
1. Direct Congruence: The mapping is 'Direct' when the core functional verb of the purpose of physical interaction (e.g., purpose of sliding the zipper of pencilbag is 'open') is identical or a direct synonym to the core functional verb of the digital task.
  Example: Rotating a lamp knob (purpose: adjust the lightness -> core verb: 'adjust') for an 'adjust volume' task (core verb: 'adjust') is Direct.
2. Metaphorical Congruence: For interactions that are not Direct (i.e., their core verbs do not match), the mapping is 'Metaphorical'. The association of a system effect of the interaction as analogous to the real-world effect of similar actions.
  Example: Rotating a cup (core verb: 'rotate') for an 'adjust volume' task (core verb: 'adjust'). But we could thinking that rotate the cup to control the volume is similar to rotate a volume controller like a knob. This is Metaphorical.
3. No Congruence: Any interaction that is neither Direct nor Metaphorical.
--
You must follow this logic to select the single winner:
1. Classify Congruity: First, for each candidate in the input list, classify it as 'Direct', 'Metaphorical', or 'No Congruence'.
2. Prioritize 'Direct': Check if any candidates were classified as 'Direct'.
  If YES, you will select your winner ONLY from the 'Direct' candidates. Ignore all others.
  If NO, proceed to the next step.
3. Consider 'Metaphorical': If no 'Direct' candidates exist, you will select your winner ONLY from the 'Metaphorical' candidates.
4. Metaphorical: If there is only one 'Metaphorical' candidate, it is the winner. Otherwise, output all feasible interactions.
5. Fallback: If no 'Direct' or 'Metaphorical' candidates are found, there is no winner.
6. Decision: Select only one interaction from the selection-pool based on two contextual appropriateness.
  - Task cohesion: The interaction could harmoniously integrated into the user's primary workflow with minimal disruption. Like holding the pen to stop the podcast instead of press the button
  - Ergonomics: The interaction that is physically closer to the user and require less effort to perform.
--
Task
Analyze the following list of candidate interactions for the given digital task and select the single best one.
- Context: [InputContext]
- Digital Task: [InputTask]
- Candidate Interaction List: [InputInteractionSet]
- Position Data: [InputPositionData]
--
Output
Provide your final output as a single JSON object with the following keys:
- "selection-pool": The category from which the final selection was made (e.g., "Direct").
- "winner": output the decision
- "justification": A concise, one-sentence explanation for why this interaction
--
Return the output in above format.
```

Fig. 10. Prompt For selecting appropriate interaction

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